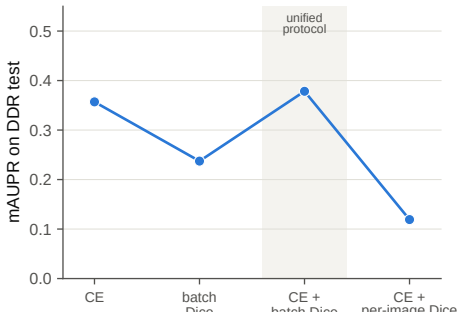
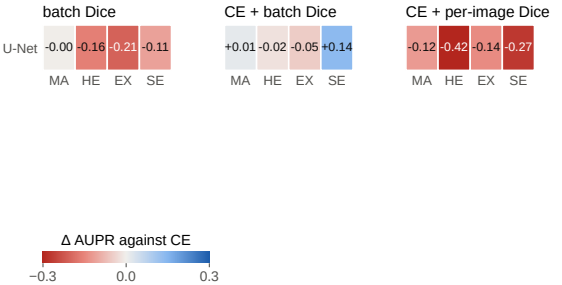
a **Loss, architecture fixed (U-Net)**



b **Per-class change against cross-entropy**



The per-image Dice term (smooth 1) of the registered U-Net recipe drives the rare-class channels to zero under patch training; the main table uses batch Dice (Methods).